

Approximating Irrational Numbers*Additional Practice 7A.02.04*

Name _____

Date _____

Period _____

1. Find the value of each. If there is no real value, write “no real value.”

$\sqrt{81}$	$\sqrt{-81}$	$\sqrt[3]{64}$	$\sqrt[3]{-64}$
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2. Why is there no real value for $\sqrt{-81}$ when there is one for $\sqrt[3]{-64}$?

3. Select ALL of the numbers below that are irrational.

☐

$\sqrt{36}$

☐

$\sqrt{40}$

☐

$\sqrt[3]{125}$

☐

$\sqrt[3]{50}$

☐

2π

☐

$\frac{3}{4}$

Questions 4 and 5: name the perfect squares or cubes first, then the root.

4. $\sqrt{95}$

a. 95 lies between the perfect squares _____ and _____.

b. So the root lies between the integers _____ and _____.

c. The root is closer to _____.

5. $\sqrt[3]{110}$

- a. 110 lies between the perfect cubes _____ and _____.
- b. So the root lies between the integers _____ and _____.
- c. The root is closer to _____.

6. Approximate $\sqrt{95}$ to the nearest tenth. Fill in the table, then decide.

Try	Result	How far from 95
$9.7 \times \text{itself}$		
$9.8 \times \text{itself}$		

To the nearest tenth, the value is _____.

7. Approximate $\sqrt[3]{110}$ to the nearest tenth. Fill in the table, then decide.

Try	Result	How far from 110
4.7 cubed		
4.8 cubed		

To the nearest tenth, the value is _____.

8. Use $\sqrt{95} \approx 9.747$ to approximate each expression to the nearest tenth.

Expression	Approximate Value
$\sqrt{95} + 3$	
$\frac{\sqrt{95}}{2}$	
$2\sqrt{95}$	

9. Approximate the value of each expression to the nearest tenth.

	Expression	Approximate Value
A	$\sqrt[3]{110}$	
B	$\pi + 2$	
C	3π	
D	$\sqrt{95}$	

10. Plot and label A, B, C and D on the number line.



11. Marcus says that because $\sqrt{7} \approx 2.6$, the value of $3\sqrt{7}$ is 7.8. His calculator says 7.9. Who is right, and what went wrong?